

## 4.10 Writing Equations of Parallel Lines

### Lesson Introduction

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|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  Benchmark           | MA.912.AR.2.3 Write a linear two-variable equation for a line that is parallel or perpendicular to a given line and goes through a given point.                                                                                                                                                                                                                                                                                                    |  Learning Targets                       | <ul style="list-style-type: none"><li>I can write the equation of a line that is parallel to a given line through a given point.</li><li>I can write the equation of a line that is parallel to a given line when given two points.</li><li>I can write the equation of a line that is parallel to a given line when given a point and the slope.</li></ul> |                                                                                                                                                                                                                                                                                            |
|  Benchmark Coherence | Building On<br>MA.8.AR.4.3 Given a mathematical or real-world context, solve systems of two linear equations by graphing.                                                                                                                                                                                                                                                                                                                          |                                                                                                                           | Working Toward<br>MA.912.GR.3.2 Given a mathematical or real-world context, use coordinate geometry to classify or justify definitions, properties and theorems involving circles, triangles, or quadrilaterals.                                                                                                                                            |                                                                                                                                                                                                                                                                                            |
|  Essential Questions | <ul style="list-style-type: none"><li>What is the relationship between two parallel lines?</li><li>How can the relationship between two parallel lines be used to write the equation of a line that is parallel to another?</li></ul>                                                                                                                                                                                                              |                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                            |
|  Lesson Narrative    | Students build on their conceptual understanding of parallel lines to begin writing the equations of those lines when given a line and point; given a line and two points; and given a line and one point and the slope. Instruction includes an exploration into how this process is the same and different for horizontal and vertical lines, as well as writing equations in all three forms (slope-intercept, point-slope, and standard form). |                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                            |
|  Component Summary | 4.10.1 Warm-Up (~5 min.)                                                                                                                                                                                                                                                                                                                                                                                                                           | Making use of patterns and structure with Legos ( <i>SE p. 214</i> )                                                      | 4.10.3 Guided Instruction (15 min.)                                                                                                                                                                                                                                                                                                                         | Making sense of the equations of parallel lines through the lens of vertical and horizontal lines and their equations ( <i>SE p. 217</i> )                                                                                                                                                 |
|                                                                                                       | 4.10.2 Collaboration (15 min.)                                                                                                                                                                                                                                                                                                                                                                                                                     | Collaborative exploration into the process of writing the equation of a line using given information ( <i>SE p. 215</i> ) | 4.10.4 Your Turn! (10 min.)                                                                                                                                                                                                                                                                                                                                 | Independent practice for writing the equation of a line in all three forms ( <i>SE p. 219</i> )                                                                                                                                                                                            |
|                                                                                                       | 4.10.5 Wrap-Up (~5 min.)                                                                                                                                                                                                                                                                                                                                                                                                                           | Cool Down for writing the equation of a parallel line in all three forms ( <i>SE p. 219</i> )                             |                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                            |
|  Resources         | Glossary                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                           | Materials                                                                                                                                                                                                                                                                                                                                                   | Additional Preparation                                                                                                                                                                                                                                                                     |
|                                                                                                       | <u>Key Terms:</u> <ul style="list-style-type: none"><li>Parallel</li></ul><br><u>Additional Terms:</u> N/A                                                                                                                                                                                                                                                                                                                                         |                                                                                                                           | <ul style="list-style-type: none"><li>N/A</li></ul>                                                                                                                                                                                                                                                                                                         | <ul style="list-style-type: none"><li>Students are not graphing or sketching parallel lines like in Lesson 9. Instead, the focus of the lesson is on the conceptual representation of the lines using an equation; therefore, straightedge tools are not needed for this lesson.</li></ul> |

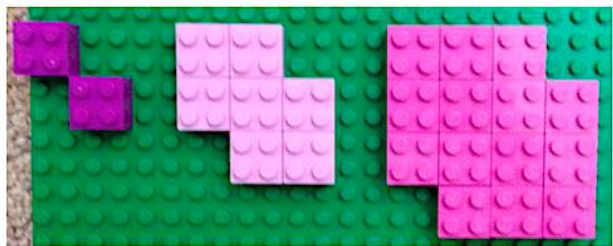
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|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>Teacher Tips               | <b>Common Misconceptions</b> <ul style="list-style-type: none"><li>Students may see any constant value as a possible <math>y</math>-intercept, therefore identifying the wrong <math>y</math>-intercept when in point-slope form.</li><li>Students may attempt to equate a vertical line with being a <math>y</math> value, therefore assuming that the equation of a vertical line would be <math>y = 3</math> instead of <math>x = 3</math>.</li></ul> | <b>Things to Consider</b> <ul style="list-style-type: none"><li>Consider the conversations you heard in Lesson 9 and how could those responses be used for learning in Lesson 10.</li><li>Additional practice for converting between forms is spiraled into Lessons 10-12, so avoid getting stuck on mastering conversions in this lesson. Monitor student understanding throughout these lessons.</li><li>Consider having the Desmos activity from Lesson 9 to project for all students to help students recall the graphs from Lesson 9.</li></ul> |
|                                                                                                                 | <b>Literacy and Language Routines</b>                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Mathematical Thinking and Reasoning (MTR) Standard</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                                 | <b>Take Turns</b><br>4.10.2 Collaboration allows students to engage in a collaborative activity on parallel lines. Give students the opportunity to try each problem in the activity, then do an agree/disagree/justify opportunity. The “Take Turns” framing helps students monitor airtime and provides equitable opportunity for engagement.                                                                                                          | <b>MA.K12.MTR.5.1: Patterns and Structure</b><br>Notice the use of patterns in making sense of mathematical concepts in 4.10.1 Warm-Up. If your classroom utilizes any strategies or Anchor Charts specifically framed through the MTR standards, use 4.10.1 Warm-Up as an opportunity to highlight those practices.                                                                                                                                                                                                                                 |
| <br>Differentiate Instruction | Provide choice for 4.10.4 Your Turn! to increase student agency. Allow students the option of whether to start with question 1 or question 2, particularly because question 1 includes a horizontal line, which could be a barrier to entry for some students.                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                                                                                 | Make informal observations during 4.10.2 Collaboration. 4.10.1 Wrap-Up is a formative assessment on writing the equation of a line, so this observation paired with informal observations from student discussions will provide a more holistic perspective on learning progress, particularly around the conceptual understanding of the meaning of parallel.                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Lesson Notes:                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## 4.10.1 Warm-Up: Lego Pattern

**Component Overview:** Students will determine visual and numeric patterns using Legos. This activity is designed to be done independently or with a partner. Consider utilizing a partner on this activity, to allow students to have an open dialogue about potentially observed patterns.



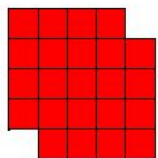
### 4.10.1 Warm-Up: Lego Pattern



1. What do you notice about the pattern?

*Answers will vary. There are more blocks with each new shape. The shape fills in more space in each row and column as the pattern progresses. Each pattern is made up of squares and the squares form squares.*

2. Draw the next set of blocks in the pattern.



3. Complete the two different ways of writing the pattern.

|                   |               |
|-------------------|---------------|
| Number of Blocks  | 2, 7, 14, 23  |
| Number of Circles | 8, 28, 56, 92 |



*MA.K12.MTR.5.1: Patterns and Structure*

*4.10.1 Warm-Up provides students with an opportunity to put patterns to work. Students connect concrete structure to abstract mathematical concepts. While you do not need to explicitly highlight this connection, students are utilizing the same skill set of applying and projecting patterns when applying the same approach to lines, graphing, and slope from Lessons 1-9 to parallel lines in this lesson.*



*If you have students in need of additional scaffolding or support before or after the lesson, consider utilizing actual Lego blocks to stimulate a kinesthetic entry-point*

## 4.10.2 Collaboration: Parallel Lines

**Component Overview:** This activity prioritizes student-centered collaboration, activating prior knowledge from Lesson 9, and extending that conceptual understanding to writing equations that represent those lines. When facilitating this activity, consider prompting questions that can continue to redirect students to build on what they know to discover what they do not know yet.

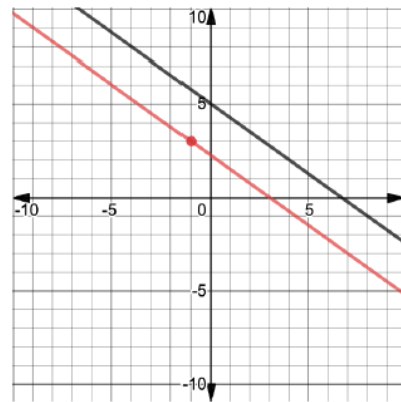


### 4.10.2 Collaboration: Parallel Lines



Lines are **parallel** if the slopes of the lines are the same.

1. The graph of the linear relationship  $y - 11 = -\frac{3}{4}(x + 8)$  is shown.



- a. Plot the point  $(-1, 3)$ . Then draw the line that passes through the point that is parallel to  $y - 11 = -\frac{3}{4}(x + 8)$ .

*Answer on graph.*

- b. What is the slope of the line that passes through the point  $(-1, 3)$ ?

$$m = -\frac{3}{4}$$

- c. Work with your partner to write the equation of the line that is parallel to  $y - 11 = -\frac{3}{4}(x + 8)$  and passes through the point  $(-1, 3)$ .

*Any equivalent equation to  $y - 3 = -\frac{3}{4}(x + 1)$*



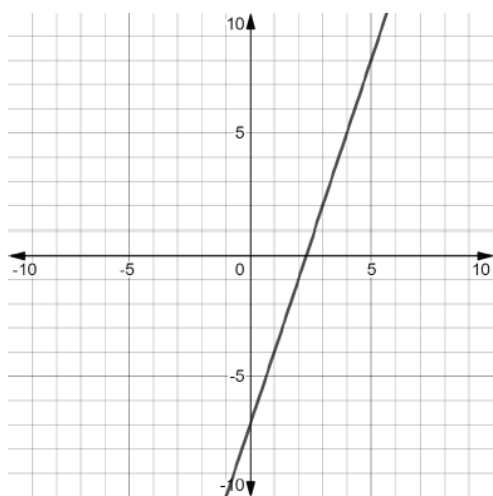
*This is not the first time that students have interacted with slope, so avoid teaching and instruction that approaches the concepts in that way. Instead, this activity is intentionally a Collaboration to facilitate student-centered learning through exploration and discovery.*



*4.10.2 Collaboration is purposefully and carefully scaffolded to provide a variety of entry-points for students to begin making sense of parallel lines. Consider offering additional support for students who need it by utilizing Math Nation's Study Expert videos from Lesson 9 that cover parallel lines (4.9.2 Exploration).*

## 4.10.2 Collaboration: Parallel Lines (continued)

2. A linear relationship is shown on the coordinate grid.



- a. Complete the equation of the line that is parallel to the line and passes through the point  $(4, -6)$ .

$$y + 6 = 3(x - 4)$$

- b. Discuss with your partner why the point-slope form of a line is the easiest form of a line to use when writing the equation of a parallel line.

*Answers will vary.*

- c. Write the equation in slope-intercept form.

$$y = 3x + (-18)$$

- d. Write the equation in standard form.

$$-3x + 1y = -18$$



Notice the point-slope form in question 2c and consider which questions might be helpful for students who are unsure where to begin.

- Where did you start when writing the new equation?
- How did you know to change those values and not the others?
- How is this similar or different than other forms of the equation of a line?

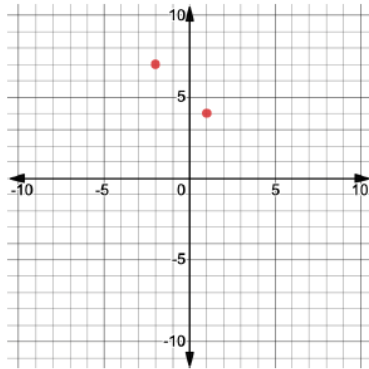


Questions 2b and 2d are discussion opportunities. Students do not need to write an explanation here, but a sample response is provided for reference.

Listen as students discuss their thinking for question 2b to determine student thinking to highlight or misconceptions to address before moving on to question 2c.

## 4.10.2 Collaboration: Parallel Lines (continued)

3. Plot the two points  $(-2, 7)$  and  $(1, 4)$  on the coordinate grid.



- a. Write the equation of the line that is parallel to the line that passes through the points  $(-2, 7)$  and  $(1, 4)$  and goes through the point  $(0, -3)$ .

$y + 3 = -x$  or any equivalent

- b. Complete the following statement about the equation of the line.  
The reason I wrote the equation in

- ☐ point-slope  
☐ slope-intercept  
☐ standard

form is because

*Answers will vary. I had the slope of  $-1$  and the point  $(0, -3)$ , so I thought that was easiest.*

- c. What method did you use to find the slope of the line?
- ☐ I used the two points  $(-2, 7)$  and  $(1, 4)$  and found  $\frac{\text{change in } y}{\text{change in } x}$ .
  - ☐ I used the graph and counted to find the  $\frac{\text{change in } y}{\text{change in } x}$ .

*Answers will vary. Either choice is correct.*



At this point in 4.10.2 Collaboration process, students should be recognizing the additional layer of finding the slope from two points. While circulating during 4.10.2 Exploration, listen for student talk and consider prompting questions to create entry-points for students who are stuck.

- What information do you know?
- How is this the same and/or different than what you had in question 1 or 2?
- What does that make you think about what you need to do or find?



Notice the sentence stems included as scaffolding for students making sense of their rationale in questions 3b and 3c. Listen for students' verbal processing of their rationale, and consider highlighting any common misconceptions or strong mathematical thinking based on student responses here.



## 4.10.3 Guided Instruction: Parallel Lines

**Component Overview:** Students begin synthesizing their findings with all three forms and explore what that means in the context of horizontal and vertical lines.



### 4.10.3 Guided Instruction: Parallel Lines

1. Write the equation of the line, in all three forms, that is parallel to the line that passes through the points  $(-5, -2)$  and  $(-3, 2)$  and goes through the point  $(7, -1)$ .

| Point-Slope Form   | Slope-Intercept Form | Standard Form                          |
|--------------------|----------------------|----------------------------------------|
| $y + 1 = 2(x - 7)$ | $y = 2x - 15$        | $2x - y = 15$<br>or<br>$-2x + y = -15$ |

2. Write the equation of the line that passes through  $(4, -7)$  and is parallel to  $y = -1$ .  
 $y = -7$
3. Discuss with your partner how to remember that the slope of a horizontal line is 0.  
*A horizontal line has a slope of 0 because the change in  $y$  is 0 between any two points on the horizontal line. 0 divided by any change in  $x$  is 0.*
4. Write the equation of the line that passes through  $(4, -7)$  and is parallel to the line that passes through the points  $(-1, -2)$  and  $(-1, 6)$ .  
*A line parallel to  $x = -1$  is  $x = 4$*
5. With your partner, discuss how to remember that the slope of a vertical line is undefined.  
*A vertical line has a slope that is undefined because the change in  $x$  is 0 between any two points on the vertical line. Any change in  $y$  divided by 0 is undefined.*



Students have plenty of practice with writing equations and converting between forms throughout Lessons 10-12. Consider prioritizing the discussion on horizontal and vertical lines in question 2 in this activity rather than spending too much time on question 1.



Lesson 9 gave students the opportunity to compare the difference between a slope of 0 and an undefined slope. questions 2b and 2d seek to activate that prior knowledge to build toward writing their equations. Consider the following prompts for facilitating discussion:

- Why is 0 the slope of a horizontal line? Prove mathematically.
- Why is the slope of a vertical line undefined? Prove mathematically.
- What does that make you think about what their equations might look like? What would be missing?

## 4.10.3 Guided Instruction: Parallel Lines (continued)

6. Complete the table.

| Type            | Sketch | Slope     | Equations of Parallel Lines    |
|-----------------|--------|-----------|--------------------------------|
| Horizontal Line |        | 0         | $y = \underline{\hspace{1cm}}$ |
| Vertical Line   |        | Undefined | $x = \underline{\hspace{1cm}}$ |

7. Write the equation of the line, in all three forms, that is parallel to the line  $-12x + 15y = 45$  and goes through the point  $(-9, 14)$ .

| Point-Slope Form              | Slope-Intercept Form               | Standard Form    |
|-------------------------------|------------------------------------|------------------|
| $y - 14 = \frac{4}{5}(x + 9)$ | $y = \frac{4}{5}x + \frac{106}{5}$ | $-4x + 5y = 106$ |



The first question in 4.10.4 Your Turn! utilizes the equation of a vertical line. Visit that question to consider what prompts to use to begin framing students to successfully engage with that question.



Notice the use of fractions when converting to slope-intercept and standard form. Consider showing an On-Ramp video on operations with fractions to provide scaffolding for adding, subtracting, and multiplying fractions.



## 4.10.4 Your Turn!

**Component Overview:** Students work independently to practice writing the equation of a line parallel to another. Notice the similarity between 4.10.4 Your Turn! and 4.10.5 Wrap-Up to effectively monitor and provide intervention for students before 4.10.5 Wrap-Up.



### 4.10.4 Your Turn!

- Write the equation of the line that passes through  $(-3, 8)$  and is parallel to  $x = 4$ .

$$x = -3$$

- Write the equation of the line, in all three forms, that is parallel to the line  $14x - 2y = 51$  and goes through the point  $(3, -11)$ .

| Point-Slope Form    | Slope-Intercept Form | Standard Form   |
|---------------------|----------------------|-----------------|
| $y + 11 = 7(x - 3)$ | $y = 7x - 32$        | $-7x + y = -32$ |



Consider giving students agency and choice in deciding which question to start with. Students who might get frustrated and stuck on question 1 because “it looks different” might find more success by starting on question 2, and then being prompted to connect how question 1 is similar or different.

- How is question 1 the same? How is it different?
- Is 4 the slope? Why or why not?
- If 4 is not the slope, then what does 4 represent?
- So, what is the slope?
- Now that you have slope and a point, what's next?

## 4.10.5 Wrap-Up: Cool Down

**Component Overview:** Students apply their learning for an independent exit ticket.



### 4.10.5 Wrap-Up: Cool Down

- Write the equation of the line, in all three forms, that is parallel to the line that passes through the points  $(-6, 8)$  and  $(-2, -4)$  and goes through the point  $(3, -5)$ .

| Point-Slope Form    | Slope-Intercept Form | Standard Form |
|---------------------|----------------------|---------------|
| $y + 5 = -3(x - 3)$ | $y = -3x + 4$        | $3x + y = 4$  |



Lesson 11 follows a similar approach to perpendicular lines, and Lesson 12 further explores parallel and perpendicular lines. Consider using data from this Wrap-Up to inform small group or intervention strategies for Lessons 11-12.